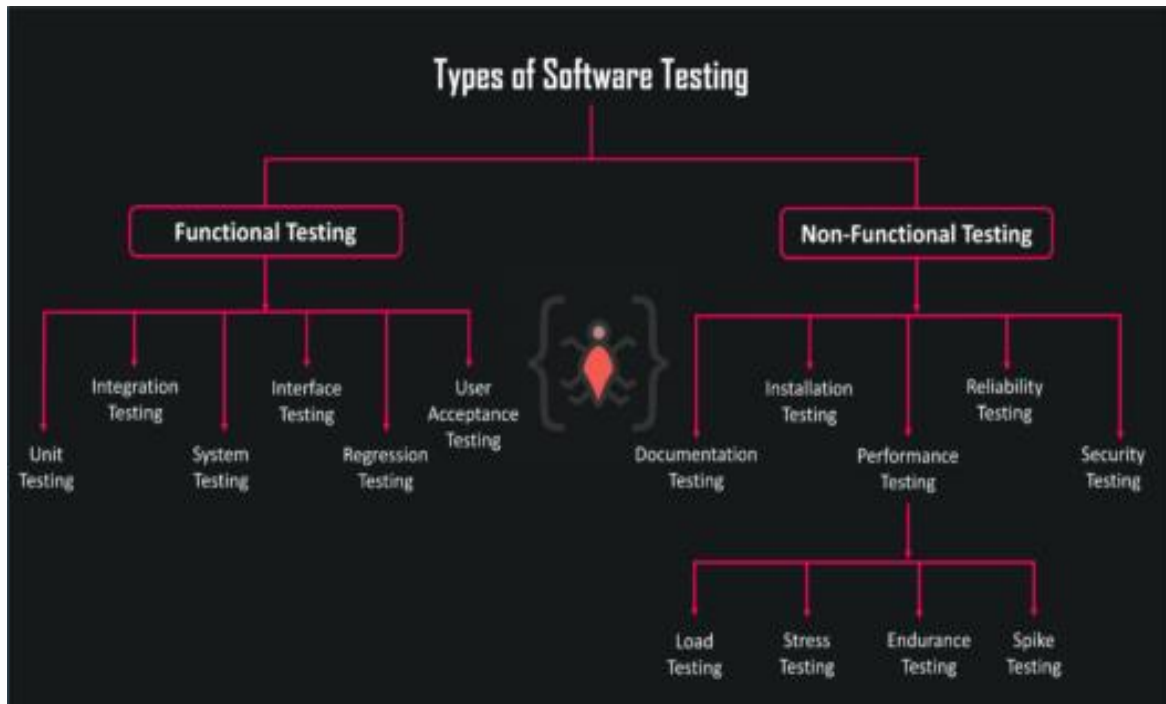




Alpha testing

Alpha testing is a kind of testing to look for all the errors and issues in the entire software. This kind of test is done at the last phase of app development and is performed at the place of the developers, before launching the product or before delivering it to the client to ensure that the user/client gets an error-free software application.

Alpha testing is run before the beta testing, which means that after performing alpha testing, you need to run beta testing.

Alpha testing is not performed in the real environment. Rather, this kind of tests is done by creating a virtual environment that resembles a real environment.

Acceptance testing

The client who will purchase your software will perform acceptance testing (also known as User Acceptance Testing) to see if the software can be accepted or not by checking whether your software meets all the client's requirements and preferences. If your

software doesn't meet all the requirements or if your client doesn't like something in the app, they may request you to make changes before accepting the project.

Ad-hoc testing

As the name suggests, ad-hoc testing is a kind of testing that is performed in an ad-hoc manner, without using any test cases, plans, documentation, or systems. Unlike all other types of testing, this kind of testing is not carried out in a systematic manner.

Although finding errors can be difficult without using test cases, there are technical issues that are easily detected through an ad-hoc test, but are hard to find through other testing approaches that use test cases.

This informal type of software testing can be executed by any person involved with the project.

Accessibility testing

Testing whether your software is accessible to disabled people or not is termed as accessible testing. For this type of tests, you need to check if disabled people such as those who are color blind, blind, and deaf can use your application.

The right choice of color and contrast need to be made to make your software accessible to color-blind people.

Beta Testing

Beta Testing is a formal type of Software Testing which is carried out by the customer. It is performed in **the Real Environment** before releasing the product to the market for the actual end-users.

Beta Testing is carried out to ensure that there are no major failures in the software or product and it satisfies the business requirements from an end-user perspective. Beta Testing is successful when the customer accepts the software.

Usually, this testing is typically done by end-users or others. It is the final testing done before releasing an application for commercial purpose. Usually, the Beta version of the software or product released is limited to a certain number of users in a specific area.

So end-user actually uses the software and shares the feedback to the company. Company then takes necessary action before releasing the software to the worldwide.

Backward compatibility testing

Backward compatibility testing is carried out to test if a brand new or an updated version of an application is compatible with the previous versions of the environments (such as operating systems and web browsers) on which the software runs. Sometimes, some application is updated specifically to match the standard and style of a newer, more modern environment. In that case, support for backward compatibility is necessary.

Backward compatibility testing ensures that all those who are using the older versions of a particular environment can use your software.

Black box testing

Performed by the QA team of a company, black box testing is a testing technique that involves the checking of the application's functionality without having any technical knowledge of the application, like the knowledge of the code's logic, how the code works, knowledge of the internal structure, etc.

Boundary Value Testing

Boundary Value Testing is performed for checking if defects exist at boundary values. Boundary Value Testing is used for testing a different range of numbers. There is an upper and lower boundary for each range and testing is performed on these boundary values.

If testing requires a test range of numbers from 1 to 500 then Boundary Value Testing is performed on values at 0, 1, 2, 499, 500 and 501.

Comparison Testing

Testing technique which compares the product strengths and weaknesses with previous versions or other similar products. Can be performed by tester, developers, product managers or product owners.

Compatibility testing

Compatibility testing involves compatibility checking of the software with different operating systems, web browsers, network environments, hardware, and so on. It

checks whether the developed software application is working fine with different configurations.

To give you a few examples, if the software is a Windows app, it should be checked whether it is compatible with different versions of the Windows operating system. If it's a web application, it is tested whether the app is easily accessible from different versions of the widely-used web browsers. And if it's an Android app, it should be checked whether it is working well with all the commonly used versions of the Android operating system.

Configuration Testing

Testing technique which determines minimal and optimal configuration of hardware and software, and the effect of adding or modifying resources such as memory, disk drives and CPU. Usually it is performed by the Performance Testing engineers.

Compliance Testing

Type of testing which checks whether the system was developed in accordance with standards, procedures and guidelines. It is usually performed by external companies which offer "Certified OGC Compliant" brand.

Domain Testing

White box testing technique which contains checkings that the program accepts only valid input. It is usually done by software development teams and occasionally by automation testing teams.

Equivalence Partitioning

It is a testing technique and a type of Black Box Testing. During this Equivalence Partitioning, a set of the group is selected and a few values or numbers are picked up for testing. It is understood that all values from that group generate the same output.

The aim of this testing is to remove redundant test cases within a specific group which generates the same output but not any defect.

Suppose, the application accepts values between -10 to +10 so using equivalence partitioning the values picked up for testing are zero, one positive value, one negative value. So the Equivalence Partitioning for this testing is -10 to -1, 0, and 1 to 10.

Functional Testing

This type of testing ignores the internal parts and focuses only on the output to check if it is as per the requirement or not. It is a Black-box type testing geared to the functional requirements of an application.

Fuzz Testing

Software testing technique that provides invalid, unexpected, or random data to the inputs of a program - a special area of mutation testing. Fuzz testing is performed by testing teams.

Graphical User Interface (GUI) Testing

The objective of this GUI Testing is to validate the GUI as per the business requirement. The expected GUI of the application is mentioned in the Detailed Design Document and GUI mockup screens.

The GUI Testing includes the size of the buttons and input field present on the screen, alignment of all text, tables, and content in the tables.

It also validates the menu of the application, after selecting different menu and menu items, it validates that the page does not fluctuate and the alignment remains same after hovering the mouse on the menu or sub-menu.

Load Testing

It is a type of Non-Functional Testing and the objective of Load Testing is to check how much load or maximum workload a system can handle without any performance degradation.

By running load tests, we can know the capacity of taking load of a system.

Monkey Testing

Monkey Testing is carried out by a tester assuming that if the monkey uses the application then how random input, values will be entered by the Monkey without any knowledge or understanding of the application.

The objective of Monkey Testing is to check if an application or system gets crashed by providing random input values/data.

Monkey Testing is performed randomly and no test cases are scripted and it is not necessary to be aware of the full functionality of the system.

Performance testing

Performance tests check the behaviors of the system when it is under significant load. These tests are non-functional and can have the various form to understand the reliability, stability, and availability of the platform. For instance, it can be observing response times when executing a high number of requests, or seeing how the system behaves with a significant of data.

Performance tests are by their nature quite costly to implement and run, but they can help you understand if new changes are going to degrade your system.

Recovery Testing

It is a type of testing which validates how well the application or system recovers from crashes or disasters.

Recovery Testing determines if the system is able to continue the operation after a disaster. Assume that application is receiving data through the network cable and suddenly that network cable has been unplugged.

Sometime later, plug the network cable; then the system should start receiving data from where it lost the connection due to network cable unplugged.

Requirements Testing

Testing technique which validates that the requirements are correct, complete, unambiguous, and logically consistent and allows designing a necessary and sufficient set of test cases from those requirements. It is performed by QA teams.

Security testing

Security tests are performed to ensure the security of your application, in order that security breaches can be prevented. Security experts run this kind of tests to see how much your software is secure from attacks and to find security issues so that the app's security can be strengthened.

White Box Testing

White Box Testing is based on the knowledge about the internal logic of an application's code.

It is also known as Glass box Testing. Internal software and code working should be known for performing this type of testing. Under these tests are based on the coverage of code statements, branches, paths, conditions, etc.

Volume Testing

Volume Testing is a type of Non-Functional Testing performed by the Performance Testing team.

The software or application undergoes a huge amount of data and Volume Testing checks the system behavior and response time of the application when the system came across such a high volume of data. This high volume of data may impact the system's performance and speed of the processing time.

Unit Testing

Testing of an individual software component or module is termed as Unit Testing. It is typically done by the programmer and not by testers, as it requires detailed knowledge of the internal program design and code. It may also require developing test driver modules or test harnesses.

Stress Testing

This testing is done when a system is stressed beyond its specifications in order to check how and when it fails. This is performed under heavy load like putting large number beyond storage capacity, complex database queries, continuous input to the system or database load.

Smoke Testing

This test is done to make sure that software under testing is ready or stable for further testing .

It is called smoke test as testing initial pass is done to check if it did not catch the fire or smoked in the initial switch on.

Example:

If project has 2 modules so before going to module
make sure that module 1 works properly

Static Testing

Static Testing is a type of testing which is executed without any code. The execution is performed on the documentation during the testing phase.

It involves reviews, walkthrough, and inspection of the deliverables of the project. Static Testing does not execute the code instead of the code syntax, naming conventions are checked.

User Interface Testing

User interface testing involves the testing of the application's user interface. The aim of UI tests is to check whether the user interfaces have been developed according to what is described in the requirements specifications document.

By running UI tests, you can make the application's user interfaces more user-friendly and appealing to the eyes.

Usability testing

Testing the user-friendliness of an app is known as usability testing. It involves the checking of how much usable or user-friendly the app is. It is tested whether any user can easily use your software without getting stuck.

One of the best ways to test the usability of your software is to invite a few people to use your software. See if they can do certain things in your app without taking any help from you.

Scalability testing

Scalability testing verifies whether the software is scalable or not. In other words, it checks if your app performs well when the number of users, amount of data, or the number of transactions increases significantly. A software application that is not scalable may cause great business loss.

Vulnerability Testing

The testing which involves identifying weakness in the software, hardware and the network is known as Vulnerability Testing. Malicious programs, the hacker can take control of the system, if it is vulnerable to such kind of attacks, viruses, and worms.

So it is necessary to check if those systems undergo Vulnerability Testing before production. It may identify critical defects, flaws in the security.

Stability Testing

Testing technique which attempts to determine if an application will crash. It is usually conducted by the performance engineer.

System Testing

It is a level of software testing where a complete and integrated software is tested. The purpose of this test is to evaluate the system's compliance with the specified requirements.

Reliability testing

Reliability testing is a type of software testing that verifies if the software is reliable or not. In other words, it checks whether the software runs error-free and that one can rely on it.

For example, if a user's important information stored in the database of the software gets suddenly deleted after a few months because of some error in the code, we can say that the software is not reliable.

Recovery testing

Recovery testing involves the checking of whether the application can recover from crashes and how well it recovers. In this kind of tests, testers observe how well the software can come back to the normal flow of execution. Crashes can happen anytime.

Even if your software is of exceptional quality, crashes may happen. You don't know when they may take place and annoy the users.

So you have to implement mechanisms that will recover the software application quickly and that will make the application run smoothly again.

API testing

Just like unit testing, API testing is also a code-level testing type. The basic difference between unit testing and API testing is that unit testing is performed by the development team whereas API testing is handled by the QA team.

Regression Testing

Every time new module is added leads to changes in program. This type of testing make sure that whole component works properly even after adding components to the complete program.

Performance Testing

Performance Testing is defined as a type of software **testing** to ensure software applications will perform well under their expected workload.

Integration Testing

It is a level of software testing where individual units are combined and tested as a group. The purpose of this level of testing is to expose faults in the interaction between integrated units.

Approaches of Integration Testing

- **Big Bang** is an approach to Integration Testing where all or most of the units are combined together and tested at one go. This approach is taken when the testing team receives the entire software in a bundle.

- **Top Down** is an approach to Integration Testing where top-level units are tested first and lower level units are tested step by step after that. This approach is taken when top-down development approach is followed.
- **Bottom Up** is an approach to Integration Testing where bottom level units are tested first and upper-level units step by step after that. This approach is taken when bottom-up development approach is followed.
- **Sandwich/Hybrid** is an approach to Integration Testing which is a combination of Top Down and Bottom Up approaches.

Gorilla Testing

Gorilla Testing is a testing type performed by a tester and sometimes by the developer as well. In Gorilla Testing, one module or the functionality in the module is tested thoroughly and heavily. The objective of this testing is to check the robustness of the application.

Equivalence Partitioning

It is a testing technique and a type of Black Box Testing. During this Equivalence Partitioning, a set of the group is selected and a few values or numbers are picked up for testing. It is understood that all values from that group generate the same output. The aim of this testing is to remove redundant test cases within a specific group which generates the same output but not any defect.

Suppose, the application accepts values between -10 to +10 so using equivalence partitioning the values picked up for testing are zero, one positive value, one negative value. So the Equivalence Partitioning for this testing is -10 to -1, 0, and 1 to 10.

Browser compatibility testing

As the name says, browser compatibility testing checks a web application for browser compatibility. More specifically, it is tested whether the web app can easily be accessed from all versions of the major web browsers.

It is a specific form of compatibility testing, while compatibility testing checks for general compatibility.